

SHAW AFB, SC, USA

WMO: 747900

Lat: 33.967N

Lon: 80.467W

Elev: 74

StdP: 100.45

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13849

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.8	-2.8	-14.5	1.1	0.2	-11.9	1.4	1.9	10.5	12.2	9.2	11.4	2.4	30	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	35.7	24.3	34.3	24.2	33.0	23.9	26.2	32.3	25.6	31.5	25.1	30.7	3.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	19.8	28.4	24.1	19.1	27.7	23.7	18.6	27.3	81.3	32.3	78.8	31.6	76.7	30.7	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.8	7.0	DB	-8.9	37.5	1.9	1.7	-10.3	38.7	-11.4	39.7	-12.5	40.7	-13.9	41.9	(4)
(5)			WB	-9.7	27.3	2.0	0.7	-11.1	27.8	-12.3	28.2	-13.4	28.5	-14.9	29.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.0	7.5	9.6	13.3	17.8	22.4	26.0	27.4	26.8	24.0	18.5	12.5	9.4	(6)	
	DBStd	8.13	5.68	5.22	5.03	4.06	3.46	2.66	1.99	2.23	2.89	4.08	4.52	5.10	(7)	
	HDD10.0	311	119	66	26	2	0	0	0	0	0	1	24	74	(8)	
	HDD18.3	1328	338	246	167	58	9	1	0	0	2	49	180	279	(9)	
	CDD10.0	3227	41	56	129	237	383	480	540	521	421	263	101	56	(10)	
	CDD18.3	1202	1	3	12	43	133	230	282	263	172	53	6	3	(11)	
	CDH23.3	11276	7	23	131	431	1227	2226	2853	2529	1383	409	44	13	(12)	
(13)	CDH26.7	4771	0	2	20	111	470	1019	1359	1156	536	96	2	0	(13)	
(14)	Wind	WSAvg	3.1	3.3	3.5	3.6	3.6	3.3	3.0	2.9	2.7	2.9	2.8	3.0	(14)	
(15) Precipitation	PrecAvg	1193	96	89	97	78	88	133	128	137	100	90	69	86	(15)	
	PrecMax	1521	207	186	186	164	181	265	305	245	207	374	177	166	(16)	
	PrecMin	905	33	33	16	24	33	45	58	65	4	0	5	20	(17)	
	PrecStd	196	52	40	43	38	37	54	61	57	48	76	45	43	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.1	26.2	28.8	31.2	35.0	37.0	37.3	37.3	34.5	31.2	27.0	25.1	(19)	
		MCWB	17.2	17.9	18.3	20.0	22.2	24.1	24.2	24.6	22.9	23.1	19.7	18.9	(20)	
	2%	DB	21.3	23.4	26.9	29.5	32.7	34.9	35.5	35.2	32.9	29.1	24.7	22.6	(21)	
		MCWB	16.0	16.3	17.4	19.1	21.7	24.1	24.6	24.8	23.1	21.3	17.7	17.7	(22)	
	5%	DB	19.0	21.2	24.8	27.8	31.2	33.4	34.1	33.7	31.5	27.5	22.6	20.3	(23)	
		MCWB	14.8	14.6	16.2	18.4	21.4	23.7	24.5	24.5	22.8	20.1	16.7	16.8	(24)	
	10%	DB	16.8	18.7	22.3	26.0	29.5	32.1	32.8	32.3	30.0	25.7	20.5	18.0	(25)	
		MCWB	13.4	13.7	15.1	17.6	20.9	23.4	24.5	24.3	22.4	19.3	16.3	15.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.3	19.3	20.2	22.4	24.2	26.4	27.1	26.9	25.5	24.6	21.3	20.8	(27)	
		MCDB	21.2	23.5	24.8	27.3	29.9	32.9	33.6	33.4	30.5	28.0	23.6	22.6	(28)	
	2%	WB	17.6	18.0	19.0	21.2	23.4	25.5	26.2	26.2	24.7	23.4	19.9	19.1	(29)	
		MCDB	19.9	21.9	23.8	25.9	29.2	32.0	32.3	32.2	29.5	26.9	22.2	21.2	(30)	
	5%	WB	15.9	16.4	18.0	20.2	22.7	24.9	25.6	25.6	24.1	22.3	18.7	17.6	(31)	
		MCDB	18.2	19.6	22.1	25.0	28.3	31.1	31.6	31.1	28.7	25.4	21.4	19.2	(32)	
	10%	WB	13.9	14.7	16.8	19.2	21.9	24.2	25.1	25.1	23.5	20.9	17.2	15.8	(33)	
		MCDB	16.3	17.8	20.5	24.0	27.1	29.9	30.8	30.2	27.6	24.2	19.9	18.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	12.2	13.0	13.0	11.9	10.8	10.5	10.3	10.4	11.8	12.4	11.3	(35)	
		MCDBR	13.7	15.0	15.4	14.8	13.5	12.3	12.0	12.0	12.2	13.0	14.1	13.2	(36)	
		MCWBR	8.9	9.0	7.4	6.3	4.7	4.0	3.8	3.6	4.2	6.0	8.2	8.8	(37)	
	5% WB	MCDBR	11.6	13.3	12.7	12.1	11.4	11.1	10.7	10.6	9.9	10.0	11.0	11.2	(38)	
		MCWBR	8.8	9.5	7.3	6.4	4.6	4.2	4.0	3.7	3.9	5.5	7.9	8.7	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.320	0.329	0.364	0.409	0.473	0.505	0.552	0.521	0.438	0.394	0.344	0.316	(40)	
	taud		2.499	2.471	2.398	2.262	2.131	2.085	1.978	2.056	2.275	2.365	2.466	2.548	(41)	
	Ebn at Noon		893	922	910	879	823	793	754	771	824	835	854	878	(42)	
	Edh at Noon		85	97	113	135	156	163	181	164	126	107	87	77	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.78	3.41	4.50	5.66	6.15	6.40	6.10	5.53	4.75	3.98	3.06	2.44	(44)	
	RadStd		0.15	0.36	0.34	0.40	0.48	0.45	0.38	0.27	0.41	0.48	0.24	0.22	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+177	+134	(46)	
(47)	Regional (64 neighbors)	+0.55	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+190	+135	(47)	

Nomenclature: See separate page